Economia das Organizações e dos Incentivos — MPE Insper

Exercise List: Real Book Exercises, Lecture 5

How to use this list. Every exercise below is a real exercise from one of the course books, reproduced verbatim: the source line gives author, book, chapter, exercise, the printed page and, in brackets, the page your PDF viewer shows for the file in `Livros/`, and whether the source gives the answer. Muñoz-Garcia & Toro-Gonzalez print a worked answer right after each exercise; MWG answers are in the MWG solutions manual in `Livros/`; Tadelis, Gibbons and Cabral give no answer in the book. When an exercise refers to a model, game or figure stated elsewhere in its book, that material is copied into the same block under "Referenced material", so every exercise can be solved where it is printed. There is one block per lecture of the real schedule. Each block has ten exercises in strictly rising difficulty, rated from 1 (apply a definition) to 5 (graduate level) next to each heading: three foundations, five exam level, two advanced. Try each exercise in the space provided before opening the book's answer.

Difficulty path. Lecture 5: from unraveling (1) through punishments of finite length, carrot and stick and the both-horizons bar game (Tadelis 10.7) to Gibbons 2.20 (5), a proof that Rubinstein's stationary strategies are a Nash equilibrium and are subgame-perfect, checking every history of an infinite game.

Block	#	Source	p. [PDF]	Level	What it trains	Diff.	Answer
L5	1 (41)	Cabral 7.13†	182 [194]	Found.	Finitely repeated PD unravels	1	none
	2 (42)	Tadelis 10.2†	215 [234]	Found.	Grim trigger; threshold δ for two paths	2	none
	3 (43)	Muñoz-Garcia Ch. 6 Ex. 1‡	274 [285]	Found.	One-, two- and ten-period punishments	2	book
	4 (44)	Muñoz-Garcia Ch. 6 Ex. 9†	305 [316]	Exam	Feasible, individually rational payoffs; folk theorem	3	book
	5 (45)	Tadelis 10.3‡	215–216 [234–235]	Exam	Length-T punishments: δ_1 vs. δ_2	3	none
	6 (46)	Tadelis 9.1	186–187 [205–206]	Exam	Stage game played twice: all SPE, carrot and stick	4	none
	7 (47)	Gibbons 2.11‡	135 [142]	Exam	A non-Nash first-stage payoff in a twice-played game	4	none
	8 (48)	Tadelis 11.5	236–237 [255–256]	Exam	Three periods with $\delta_1 \neq \delta_2$	4	none
	9 (49)	Tadelis 10.7‡	217–218 [236–237]	Adv.	Quality: ten days vs. forever, in parameters	4	none
	10 (50)	Gibbons 2.20†	137–138 [144–145]	Adv.	Rubinstein strategies are an SPE (one-stage deviations)	5	none

#: item number in this lecture's list, then its number in the combined list (`lista-reais-lectures-01-05.pdf`) in brackets. Items not in the list for Lectures 1–4 (`lista-reais-lectures-01-04.tex`) are marked: † added in the 30-item version, ‡ added in the 50-item version. Every other item is reused from that list with its statement unchanged, some in a different block or position. "Diff.": difficulty from 1 (apply a definition) to 5 (graduate level). Pages: printed page, then the PDF viewer page in brackets (offsets in `Livros/book-registry.json`; MWG is a two-pages-per-sheet scan). "Answer": *book* = worked answer printed after the exercise; *manual* = MWG solutions manual in `Livros/`; *none* = no answer in the book. Books: Tadelis (2013), *Game Theory: An Introduction*; Gibbons (1992), *Game Theory for Applied Economists*; Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed.; Mas-Colell, Whinston & Green (1995), *Microeconomic Theory* (MWG); Cabral (2017), *Introduction to Industrial Organization*, 2nd ed.

Block 5 — Repeated games and bargaining *(Lecture 5)*

Exercise 1. (item 41 of the combined list) *Finitely repeated game* [Foundations; difficulty 1/5]

Source. Cabral (2017), *Introduction to Industrial Organization*, 2nd ed., Ch. 7 "Games and Strategies", Exercises, Exercise 7.13 "Finitely repeated game", p. 182 [PDF 194]. No answer in the book. Reproduced verbatim.

7.13. Finitely repeated game. Suppose that the game depicted in Figure 7.1 is repeated T times, where T is known. Show that the only subgame perfect equilibrium is for players to choose (B, R) in every period.

Referenced material. Source. "the game depicted in Figure 7.1": Cabral (2017), Ch. 7, Figure 7.1 "The 'prisoner's dilemma' game" and the sentence that explains how its cells are read, p. 160 [PDF 172]. Figure cropped from the page; text reproduced verbatim.

For each combination of strategies by each player, the respective matrix cell shows the payoffs received by each player. In the lower left corner, the payoff received by Player 1; in the top right corner, the payoff received by Player 2.

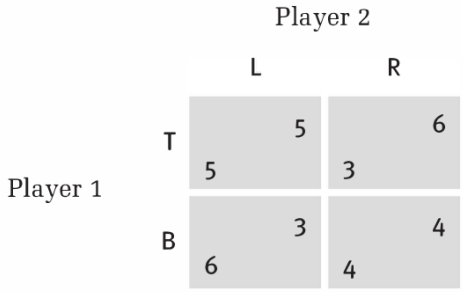


FIGURE 7.1
The "prisoner's dilemma" game

Resolution:

Exercise 2. (item 42 of the combined list) *Grim Trigger* [Foundations; difficulty 2/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 10 "Repeated Games", Exercise 10.2 "Grim Trigger", p. 215 [PDF 234]. No answer in the book. Reproduced verbatim.

10.2 Grim Trigger: Consider the infinitely repeated game with discount factor $\delta < 1$ of the following variant of the Prisoner's Dilemma:

		Player 2		
		L	C	R
Player 1	T	6, 6	-1, 7	-2, 8
	M	7, -1	4, 4	-1, 5
	B	8, -2	5, -1	0, 0

a. For which values of the discount factor δ can the players support the pair of actions (M, C) played in every period?

b. For which values of the discount factor δ can the players support the pair of actions (T, L) played in every period? Why is your answer different from that for (a)?

Resolution:

Exercise 3. (item 43 of the combined list) *Infinitely Repeated Prisoner's Dilemma Game*

[Foundations; difficulty 2/5]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 6 "Repeated Games and Correlated Equilibria", Exercise 1 "Infinitely Repeated Prisoner's Dilemma Game"[A], p. 274 [PDF 285]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only); Table 6.1 rebuilt from the same page.

Consider the infinitely repeated version of the following prisoner's dilemma game, where C denotes confess and NC represents not confess (Table 6.1):

Part (a) Can players support the cooperative outcome (NC, NC) as a SPNE by using tit-for-tat strategies, whereby players punish deviations from NC in a past period by reverting to the stage-game Nash equilibrium where they both confess (C, C) for just one period, and then returning to cooperation?

Part (b) Consider now that players play tit-for-tat, but reverting to the stage-game Nash equilibrium where they both confess (C, C) for two periods, and then players return to cooperation. Can the cooperative outcome (NC, NC) be supported as a SPNE?

Part (c) Suppose that players use strategies that punish deviation from cooperation by reverting to the stage-game Nash equilibrium for ten periods before returning to cooperation. Compute the threshold discount factor, δ, above which cooperation can be sustained as a SPNE of the infinitely repeated game.

		Player 2	
		C	NC
Player 1	C	2, 2	6, 0
	NC	0, 6	3, 3

Table 6.1 Prisoner's dilemma game

Resolution:

Exercise 4. (item 44 of the combined list) *Representing Feasible, Individually Rational Payoffs* [Exam level; difficulty 3/5]

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, 2nd ed., Ch. 6 "Repeated Games and Correlated Equilibria", Exercise 9 "Representing Feasible, Individually Rational Payoffs"[C], p. 305 [PDF 316]. Worked answer in the book, right after the statement. Reproduced verbatim (statement only); Table 6.6, printed inside the answer on the same page, rebuilt.

Consider an infinitely-repeated game where the stage game is depicted Table 6.6. Player 1 chooses rows and player 2 chooses columns. Players discount the future using a common discount factor δ.

Part (a) What outcomes in the stage-game are consistent with Nash equilibrium play?

Part (b) Let v_1 and v_2 be the repeated game payoffs to player 1 and player 2 respectively. Draw the set of feasible payoffs from the repeated game, explaining any normalization you use.

Part (c) Find the set of individually rational feasible set of payoffs.

Part (d) Find a Nash equilibrium in which the players obtain (9, 9) each period. What restrictions on δ are necessary?

		Player 2	
		L	R
Player 1	U	9, 9	1, 10
	D	10, 1	7, 7

Table 6.6 Simultaneous-move game

Resolution:

Exercise 5. (item 45 of the combined list) *Not-So-Grim Trigger* [Exam level; difficulty 3/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 10 "Repeated Games", Exercise 10.3 "Not-So-Grim Trigger", pp. 215–216 [PDF 234–235]. No answer in the book. Reproduced verbatim, with the book's footnote 12 (p. 216). Here T is the length of the punishment.

10.3 Not-So-Grim Trigger: Consider the infinitely repeated Prisoner's Dilemma with discount factor $\delta < 1$ described by the following matrix:

		Player 2	
		m	f
Player 1	M	4, 4	-1, 5
	F	5, -1	1, 1

Instead of using grim-trigger strategies to support a pair of actions (a_1, a_2) other than (F, f) as a subgame-perfect equilibrium, assume that the players wish to choose a less draconian punishment called a "length-T punishment" strategy. If there is a deviation from (a_1, a_2) then the players will play (F, f) for T periods and then resume playing (a_1, a_2). Let δ_T be the critical discount factor so that if $\delta > \delta_T$ then the adequately defined strategies will implement the desired path of play with length-T punishment as the threat.

a. Let $T = 1$. What is the critical value δ_1 to support the pair of actions (M, m) played in every period?

b. Let $T = 2$. What is the critical value δ_T to support the pair of actions (M, m) played in every period?[12]

c. Compare the two critical values in (a) and (b). How do they differ and what is the intuition for this?

[12] Hint: You should encounter an equation of the form $a\delta^3 - (a+1)\delta + 1 = 0$, for which it is easy to see that $\delta = 1$ is a root. In this case you know that the equation can be written in the form $(\delta - 1)(a\delta^2 + a\delta - 1) = 0$, and you can solve for the other relevant root of the cubic equation.

Resolution:

Exercise 6. (item 46 of the combined list) *Multistage Practice* [Exam level; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 9 "Multistage Games", Exercise 9.1 "Multistage Practice", pp. 186–187 [PDF 205–206]. No answer in the book. Reproduced verbatim.

9.1 Multistage Practice: Consider the following simultaneous-move game that is played twice (the players observe the first-period outcome prior to the second-period play):

		Player 2		
		L	C	R
Player 1	T	10, 10	2, 12	0, 13
	M	12, 2	5, 5	0, 0
	B	13, 0	0, 0	1, 1

a. Find all the pure-strategy subgame-perfect equilibria with no discounting ($\delta = 1$). Be precise in defining history-contingent strategies for both players.

b. For each of the equilibria you found in (a), find the smallest discount factor that supports it.

Resolution:

Exercise 7. (item 47 of the combined list) *Reaching a non-Nash payoff in the first stage*

[Exam level; difficulty 4/5]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 "Dynamic Games of Complete Information", §2.6 Problems (Section 2.3), Problem 2.11, p. 135 [PDF 142]. No answer in the book (Muñoz-Garcia & Toro-Gonzalez Ch. 6 Ex. 7, p. 299, works the same game). Reproduced verbatim.

2.11. The simultaneous-move game (below) is played twice, with the outcome of the first stage observed before the second stage begins. There is no discounting. Can the payoff $(4, 4)$ be achieved in the first stage in a pure-strategy subgame-perfect Nash equilibrium? If so, give strategies that do so. If not, prove why not.

	L	C	R
T	3, 1	0, 0	5, 0
M	2, 1	1, 2	3, 1
B	1, 2	0, 1	4, 4

Resolution:

Exercise 8. (item 48 of the combined list) *Asymmetric Patience 1* [Exam level; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 11 "Strategic Bargaining", Exercise 11.5 "Asymmetric Patience 1", pp. 236–237 [PDF 255–256]. No answer in the book. Reproduced verbatim.

11.5 Asymmetric Patience 1: Consider a three-period sequential (alternating-offer) bargaining model in which two players have to split a pie worth 1 (starting with player 1 making the offer). Now the players have different discount factors, δ_1 and δ_2.

a. Compute the outcome of the unique subgame-perfect equilibrium.

b. Show that when $\delta_1 = \delta_2$ player 1 has an advantage.

c. What conditions on δ_1 and δ_2 give player 2 an advantage? Why?

Referenced material. Source. the alternating-offer bargaining model this exercise modifies ("Now the players have different discount factors"): Tadelis (2013), Ch. 11 "Strategic Bargaining", p. 221 [PDF 240]. Reproduced verbatim (excerpts); with $\delta_1 \neq \delta_2$, player i discounts with δ_i.

In the first round:

- Player 1 offers shares $(x, 1-x)$, where player 1 receives x and player 2 receives $1-x$.
- Player 2 then chooses to accept, causing the game to end with payoffs $v_1 = x$ and $v_2 = 1-x$, or reject, causing the game to move to the second round.

In the second round:

- A share $1-\delta$ of the pie is dissipated (the pie is discounted with discount factor $0 < \delta < 1$).
- Player 2 offers shares $(x, 1-x)$, where player 1 receives x and player 2 receives $1-x$.
- Player 1 chooses to accept, causing the game to end with payoffs $v_1 = \delta x$ and $v_2 = \delta(1-x)$, or reject, causing the game to move to the third round.

In the third round and beyond:

- The game continues in the same way, where following rejection in an odd period player 2 gets to offer in the next even period, and vice versa.
- Each period has further discounting of the pie, so that in period t the total pie is worth δ^{t-1}.

[. . .] Assume for now that the game ends in some prespecified final period T, which will be reached if the players did not reach an agreement in the $T-1$ periods that preceded T. Furthermore assume that period T is a "hard deadline," so that both players receive a payoff of zero if they fail to reach an agreement in this last period.

Resolution:

Exercise 9. (item 49 of the combined list) *Diluted Happiness* [Advanced; difficulty 4/5]

Source. Tadelis (2013), *Game Theory: An Introduction*, Ch. 10 "Repeated Games", Exercise 10.7 "Diluted Happiness", pp. 217–218 [PDF 236–237]. No answer in the book. Reproduced verbatim.

10.7 Diluted Happiness: Consider a relationship between a bartender and a customer. The bartender serves bourbon to the customer and chooses $x \in [0, 1]$, which is the proportion of bourbon in the drink served, while $1 - x$ is the proportion of water. The cost of supplying such a drink (standard 4-ounce glass) is cx, where $c > 0$. The customer, without knowing x, decides on whether or not to buy the drink at the market price p. If he buys the drink his payoff is $vx - p$, and the bartender's payoff is $p - cx$. Assume that $v > c$ and all payoffs are common knowledge. If the customer does not buy the drink he gets 0 and the bartender gets $-cx$. Because the customer has some experience, once the drink is bought and he tastes it, he learns the value of x, but this is only after he pays for the drink.

a. Find all the Nash equilibria of this game.

b. Now assume that the customer is visiting town for 10 days, and this "bar game" will be played on each of the 10 evenings that the customer is in town. Assume that each player tries to maximize the (nondiscounted) sum of his stage payoffs. Find all the subgame-perfect equilibria of this game.

c. Now assume that the customer is a local, and the players perceive the game as repeated infinitely many times. Assume that each player tries to maximize the discounted sum of his stage payoffs, where the discount rate is $\delta \in (0, 1)$. What is the range of prices p (expressed in the parameters of the problem) for which there exists a subgame-perfect equilibrium in which every day the bartender chooses $x = 1$ and the customer buys at the price p?

d. For which values of δ (expressed in the parameters of the problem) can the price range that you found in (c) exist?

Resolution:

Exercise 10. (item 50 of the combined list) *Rubinstein's strategies are subgame-perfect*

[Advanced; difficulty 5/5]

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 "Dynamic Games of Complete Information", §2.6 Problems (Section 2.4), Problem 2.20, pp. 137–138 [PDF 144–145]. No answer in the book. Reproduced verbatim.

2.20. Consider the following strategies in the infinite-horizon version of Rubinstein's bargaining model. (Recall the notational convention that the offer $(s, 1-s)$ means that Player 1 will get s and Player 2 will get $1-s$, independent of who made the offer.) Let $s^* = 1/(1+\delta)$. Player 1 always offers $(s^*, 1-s^*)$ and accepts an offer $(s, 1-s)$ only if $s \geq \delta s^*$. Player 2 always offers $(1-s^*, s^*)$ and accepts an offer $(s, 1-s)$ only if $1-s \geq \delta s^*$. Show that these strategies are a Nash equilibrium. Show that this equilibrium is subgame-perfect.

Referenced material. Source. "the infinite-horizon version of Rubinstein's bargaining model": Gibbons (1992), §2.1.D "Sequential Bargaining", pp. 68–70 [PDF 75–77]. Reproduced verbatim (excerpts); footnote 7 (on the discount factor as $\delta = 1/(1+r)$) omitted.

Players 1 and 2 are bargaining over one dollar. They alternate in making offers: first player 1 makes a proposal that player 2 can accept or reject; if 2 rejects then 2 makes a proposal that 1 can accept or reject; and so on. Once an offer has been rejected, it ceases to be binding and is irrelevant to the subsequent play of the game. Each offer takes one period, and the players are impatient: they discount payoffs received in later periods by the factor δ per period, where $0 < \delta < 1$.

A more detailed description of the timing of the three-period bargaining game is as follows.

(1a) At the beginning of the first period, player 1 proposes to take a share s_1 of the dollar, leaving $1-s_1$ for player 2.

(1b) Player 2 either accepts the offer (in which case the game ends and the payoffs s_1 to player 1 and $1-s_1$ to player 2 are immediately received) or rejects the offer (in which case play continues to the second period).

(2a) At the beginning of the second period, player 2 proposes that player 1 take a share s_2 of the dollar, leaving $1-s_2$ for player 2. (Note the convention that s_t always goes to player 1, regardless of who made the offer.)

(2b) Player 1 either accepts the offer (in which case the game ends and the payoffs s_2 to player 1 and $1-s_2$ to player 2 are immediately received) or rejects the offer (in which case play continues to the third period).

(3) At the beginning of the third period, player 1 receives a share s of the dollar, leaving $1-s$ for player 2, where $0 < s < 1$.

[...] Now consider the infinite-horizon case. The timing is as described previously, except that the exogenous settlement in step (3) is replaced by an infinite sequence of steps (3a), (3b), (4a), (4b), and so on. Player 1 makes the offer in odd-numbered periods, player 2 in even-numbered; bargaining continues until one player accepts an offer.

Resolution: